

LoRa Based Communication System With Gateway Intelligence And Anti Jamming Control

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Abstract—Long-range wireless communication requires extremely reliable transmission over a secure channel ideal for data sensitive environments. LoRa-based communication is implemented for an average, secured long-distance data transport. At the transmitter side, ESP32 + LoRa TRANSMITTER module is used where user enters data to be sent and before sending it AES-128 symmetric encryption is performed on data. Then the data is transmitted using LoRa module. The receiver side, which has a similar configuration of the transmitter side continuously monitors LoRa communication parameters to assess link quality. Utilizing metrics like the received signal strength indicator (RSSI) and SNR (Signal-to-Noise Ratio). When the quality of signal is weak, or interference occurs on a frequency channel by operating the direction switch the system to another channel via frequency switching and maintain stable communications. **Releability: Monitoring and visualization of the system data can be done efficiently through a cloud-based web dashboard, which provides real-time access to information along with automated cloud logging for storing communication records.** **Index Terms**—LoRa, ESP32, AES-128 Encryption, Frequency Agility, Intelligent Gateway, IoT Cloud Logging.

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I. INTRODUCTION

Long-range high-throughput wireless communication plays a crucial role in numerous real-time applications since long-distance transmission of data is required with sound security. Low Power Wide Area Networks (LPWAN) technologies which are based on LoRa have attracted much interest for long-range and low-power communication. This allows them to be applied for Internet of Things (IoT) applications such as smart cities, agricultural monitoring and remote sensing systems [1], [3], [4]. Low power is also a requirement for most sensor nodes which are battery powered and deployed in the field over long time periods – hence LoRa supports very

low power consumption. These features are particularly useful in remote areas with limited, scheduled maintenance.

Despite the advantages offered by LoRa, a number of challenges still need to be solved within the Lora-based systems with respect to data security and reliability while communicating. Wireless communication is susceptible to unauthorized access, inference on the signal and noise in nature. These factors can affect the stability of the communication link and may lead to data loss or interruption during transmission [17], [20]. In many conventional LoRa systems, gateways mainly act as passive relays that forward packets without monitoring link quality or responding to changing signal conditions [5], [18]. In addition, studies on LoRa network scalability have reported issues related to channel access, interference, and network congestion [6]–[9]. Such limitations become more noticeable when a large number of devices attempt to communicate within the same network coverage area. Communication delays and packet collisions may also increase under heavy network traffic.

To overcome these limitations, this work presents an improved LoRa communication system that combines secure data transmission with adaptive network control. AES-128 encryption is used to protect transmitted data, while an intelligent gateway continuously monitors signal quality using Received Signal Strength Indicator (RSSI) and Signal-to-Noise Ratio (SNR). Based on these values, the gateway can detect communication degradation and switch channels using a frequency agility mechanism to maintain a stable link. The system also includes a cloud-based web dashboard for real-time monitoring and cloud logging to store communication data for future analysis. Live network parameters help users quickly detect faults and evaluate system performance. By combining security, adaptive control, and monitoring, the proposed system improves the reliability of long-range wireless communication.

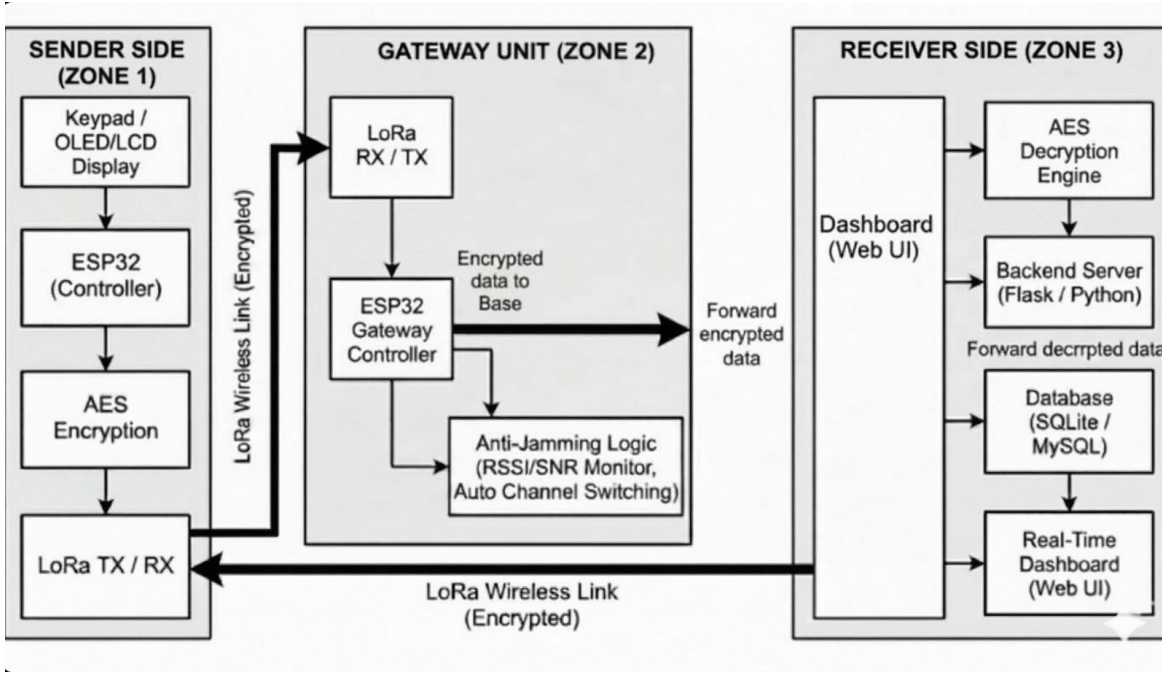


Fig. 1. System Architecture

II. LITERATURE SURVEY

Several papers have shown that AES-based security mechanisms can be employed on embedded microcontrollers to encrypt data transmitted from one device to another. Thus, such security mechanisms can be used to provide strong physical security for devices and confidentiality of transmitted data [17], [20]. These approaches provides us strong device-level security and ensure confidentiality during the long-range wireless communication.

Additionally, there are some studies devoted to the scalability and limitations of LoRa networks. For instance, research analyzing LoRaWAN architecture and operation principles emphasizes the problems of limited network capacity, interference, and collisions in the channel during communication between multiple nodes sharing the same communication channel [6]–[9]. Also, performance evaluation of LoRa modulation and network operations in an urban environment demonstrates the possibility of degradation in communication quality caused by interference and signal attenuation effects [10], [11], [16].

Several surveys have looked into the overall design and operational features of LPWAN technologies like LoRa and NB-IoT. These studies offer insights into network structure, communication protocols, and application areas of LoRa-based IoT systems [4], [5], [18]. Additionally, research has focused on improving MAC-layer optimization and spreading factor allocation to boost network efficiency and energy performance [13], [15].

Despite these advancements, many current solutions assume stable communication channels. They do not tackle problems related to signal interference or communication disruptions. In most traditional LoRa setups, the gateway acts only as a

passive relay, forwarding data packets without checking link quality. This can lead to decreased communication reliability when signal conditions worsen due to noise or interference.

To address these problems, the developed system offers a LoRa communication framework that merges AES-128 encryption with an intelligent gateway. This gateway monitors signal conditions using parameters such as Received Signal Strength Indicator (RSSI) and Signal-to-Noise Ratio (SNR). The gateway constantly detects drops in signal quality and automatically switches channels through a frequency agility mechanism to keep communication stable. A cloud-based dashboard allows for live monitoring, and data is saved for future review and analysis. This design enhances system visibility, communication stability, and data security.

III. SYSTEM ARCHITECTURE

The system architecture also supports monitoring and analysis of communication performance parameters through the cloud platform. The gateway sends the received data and signal information to the cloud server through Wi-Fi connection, while a web-based dashboard gives us the view for real-time monitoring of messages and parameters such as RSSI and SNR. All communication events and system parameters are recorded through automated cloud logging, creating a remote database that stores message history and network performance data for later analysis and evaluation.

A. Secure Node Transmission

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B. Intelligent Gateway Processing

In the second stage, we have a gateway unit which acts as an intermediate node of communication between the transmitter and the cloud platform. The gateway receives encrypted LoRa data packets and extracts the communication metadata for all the communications taking place. It extracts essential factors like RSSI and Signal-to-noise Ratio(SNR) to determine the health condition of the wireless link. With these two factors, the gateway determines whether the communication link is stable enough to forward the information to the cloud platform.

C. IoT Cloud Logging & Dashboard

System architecture also facilitates monitoring and analysis of the communication performance parameters using the cloud platform. The gateway uploads the received information on data and signals into the cloud server using the Wi-Fi connection. A web-based interface allows us to monitor the messages and performance parameters, which include RSSI and SNR. All communication activities and performance parameters are logged automatically by the cloud, thus providing a remote database of the message history and performance parameters.

IV. METHODOLOGY

In order to accomplish secure and reliable long-range communication, this methodology proposes a combination of hardware-level encryption for interference cancellation at the sink level along with cloud-based monitoring. The methodology consists of three pillars: asymmetric encryption at the source with AES-128, smart anti-jamming signal detection and counter action processing in gateway (but limited by transmission channel's error), and history cloud monitoring using IoT platform.

A. Secure Data Origination (AES-128 Protocol)

The first method of the methodology makes sure that transmitted data remains hidden from adversarial agents. The user input interface of the project is a matrix Keypad where data gets first generated. The message input is then received and sent to the ESP32 microcontroller, where it waits for specific changes by me so that it can process data ready for secure transmissions.

For confidentiality, AES with a 128-bit symmetric key is deployed in the system. AES is chosen due to its excellent security and relatively low computational implementation on embedded platforms. This scheme depends on using a common secret key by the node that is transmitting data and other nodes that are receiving encrypted information.

AES encryption of a message on the ESP32 is done in blocks and each block goes through several rounds of transformations. There are Sampling rounds such as byte substitution,

row shifting, column mixing and addition of round key. In this way, through these iterative transformations with the plaintext message being encrypted into ciphertext which can be understood without the corresponding key.

The encrypted secure ciphertext is then transmitted using a LoRa transceiver module. This is due to the low power of LoRa modulation; its design allows long-range communication with high efficiency and ensures successful data reception even in noise/radio interference prone areas.

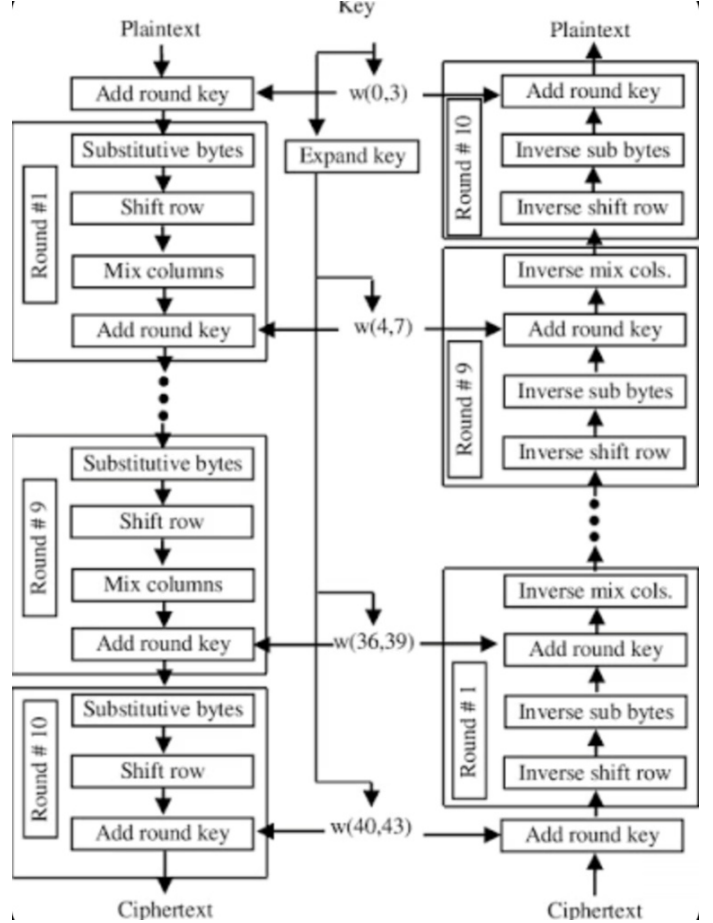


Fig. 2. AES Encryption and Decryption

B. Intelligent Mitigation (Anti-Jamming Logic)

The second stage of the methodology advances our work by ensuring robust communications through sensing and responding to signal interference. The gateway node is the central system where monitoring and decision-making process occurs in this implementation.

Gateway keeps readings of two LoRa communication parameters (Received Signal Strength Indicator(RSSI) and Signal-to-noise Ratio(SNR)) to evaluate the quality of signal constantly. RSSI values reflect the signal strength at a receiver and SNR indicates how many orders of magnitude stronger the needed signal is compared to noise from other sources in range.

These parameters are analyzed by an adaptive decision logic to decide the status of communication link. In the case that such signals happen in a signal (that is, high power for noise energy above the prescribed threshold), this means possible channel abuse / degradation and will signaled by gateways.

The gateway also initiates an adaptive frequency switching mechanism to ensure reliable communication. In this context, the gateway causes the transmitter node to switch its operation on a carrier frequency at which lower levels of interference are encountered. This synchronized channel hopping allows the communication link to circumvent an impacted frequency band and facilitate a continuation of data transfer.

C. Cloud-Based Supervision (Blynk Integration)

The last leg of the process focuses on keeping track, logging communication through a cloud-based interface. The gateway forwards this Received data with its specified signal parameters to Blynk IoT cloud platform via wifi connection. In this way, there is no need for a local server as well as laying it remote monitoring.

User interface to monitor system performance is the Blynk web dashboard. It shows transmitted messages and key communication metrics (RSSI, SNR). Users would be able to keep track of the health and reliability in real time through these indicators

In addition to visualization, the cloud platform performs automated logging of all communication events. Each transmitted message and associated signal parameter is stored in a remote database. This continuous data recording creates a permanent audit trail that can be used for performance analysis, troubleshooting, and long-term system evaluation.

V. RESULTS AND DISCUSSION

Testing this proposed system have shown successful incorporation of secure encryption, adaptable connected interference monitoring and cloud-based supervisonal. The system was verified based on testing under diverse conditions for cryptographic reliability, real time monitoring and cloud synchronization performance. The results show that this architecture retains secured data transmission while allowing reliable monitoring of the communication.

A. Cryptographic Performance and Data Integrity

The AES-128 encryption mechanism, which is implemented on the ESP32 controller to transport data securely with very low processing. All Messages From Matrix Keypad Were Encrypted Successfully At The Transmitting Node And Then Decryption Is Done Correctly Using Shared SecretKey at Receiving Side. No packets were corrupted during transmission, and the plaintext was only recovered successfully when the correct key to decrypt it had been sent. In intercept tests, the packets while they were still encrypted could not be read without a key so it seems that any data sent was perfectly unreadable which displays how well this encryption layer works in providing security for everything you are sending over.

B. Signal Analysis and Anti-Jamming Response

The gateway node tracked communication quality by using sent RSSI and SNR from the successful transmission. During the entire duration of testing under normal operating conditions, this system provided continuous communication with positive signal-to-noise ratio (SNR) levels and consistent signal strength. The gateway recognized the transmission of anomalous signals with high RSSI and low SNR when weak interferences were simulated. It has implemented upon detecting this condition the frequency switching mechanism to change communication channel into a carrier frequency with less noise. This response had a small recovery time to restore stable communication.

C. Cloud Integration and System Monitoring

Central monitoring of the communication activity was possible on Blynk IoT platform. By doing this, the gateway could send to cloud server communication data and signal parameters via WiFi. You had a web dashboard, which can visualize the messages transmitted in real time with signal metrics (RSSI SNR) Furthermore, automated log entries of system events and communications parameters to a remote database on the cloud. This feature provided fault-tolerant data storage and allowed for the constant tracking of system status metrics, detection of errors in real-time, troubleshooting issues with diagnostic logs through a root cause analysis process.

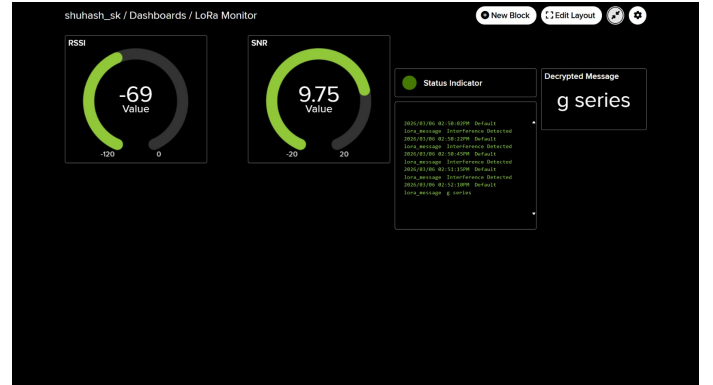


Fig. 3. IOT Dashbaord

VI. CONCLUSION AND FUTURE WORK

And the proposed ESP32 and LoRa-based communication system shows a low-power reliable application of long-range Z-Wave wireless transmission. The introduced system, in addition to providing a foundation for real time localization System at the transmitter node through integrating AES-128 encryption ensures that all data transmission is done securely and without risk of interception by third parties. As soon the user enters any input, before sending it over to API this encryption mechanism converts that information into a ciphertext which helps maintain confidentiality and ensures pristine integrity of data across the transmission path.

The system has an intelligent gateway that keeps monitoring the communication, in addition to keeping it secure. It is

performed in a continuous manner by monitoring parameters such as RSSI and SNR. In that way, by monitoring these parameters consistently, the gateway checks how healthy and reliable is the wireless link and adjusts itself in response to different signal conditions. A cloud-based monitoring dashboard to improve the visibility of system messages and communication parameters in real time as they are transmitted is also integrated. Collectively, these components turn a regular LoRa communication setup into a more secure adaptive and reliable wireless comms framework.

Upcoming upgrades in the framework would focus more for enhancing wishful thinking and anticipation features of communication network. One possible path is creating an Artificial Intelligence LoRa Environment Navigation System (AI-LENS), where the ESP32 hardware can collect detailed signal metadata such as RSSI trends, SNR variations, and packet transmission patterns. By analyzing these parameters, the system can better understand the behavior of the wireless environment and identify patterns associated with communication degradation.

The next stage of development may involve integrating lightweight machine learning techniques such as TinyML directly on the ESP32 microcontroller. This would allow the device to perform real-time pattern recognition and predict potential link degradation before it occurs. With predictive intelligence, the system could automatically adjust communication parameters such as carrier frequency or transmission power, enabling a more autonomous and adaptive communication network capable of maintaining stable connectivity in dynamic wireless environments.

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